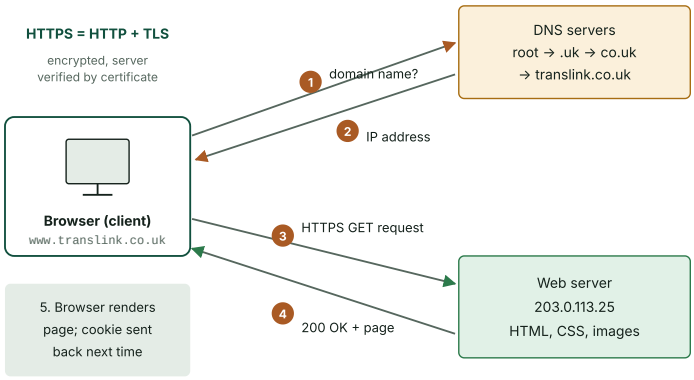
How the internet and the web work

FD16
FACT FILE

Specification: students should be able to

- Explain the terms Internet, World Wide Web (WWW), website, web page, web browser, web server, web hosting and internet service provider (ISP)
- Explain Internet Protocol (IP) addresses and how domain names are organised, including the role of the Domain Name System (DNS)
- Explain Uniform Resource Locators (URLs), HTTP, HTTPS and File Transfer Protocol (FTP)
- Explain the use of cookies, sessions and caches with the stateless HTTP protocol
- Explain how a search engine works
- Describe the role of the ITU and W3C (and the IETF) in agreeing web standards

Key words: Internet, World Wide Web (WWW), TCP/IP, IP address, IPv4 / IPv6, domain name, top-level domain (TLD), Domain Name System (DNS), internet service provider (ISP), uniform resource locator (URL), web browser, web server, HTTP / HTTPS, stateless protocol, cookie, session variable, cache, FTP, search engine



Loading a web page: the browser asks DNS for the IP address of the domain name, then sends an HTTPS request to the web server, which returns the page

Overview: the Internet and the Web

The Internet and the World Wide Web are often treated as the same thing, but they are different. The Internet is the global network of networks, linked by agreed protocols called TCP/IP so that any connected device can exchange data with any other. It grew from research networks in the 1960s and 1970s, including ARPANET in the United States, the NPL network in England, which introduced packet switching, and CYCLADES in France. Because these networks were incompatible, TCP/IP was developed as a common set of rules, and joining networks this way became known as internetworking. The World Wide Web is one of the services that runs over the Internet. Tim Berners-Lee proposed it at CERN in 1989 as a system of documents stored on servers and linked by hypertext, and CERN made it free for anyone to use in 1993. A web page is one hypertext document; a website is a group of related pages sharing a domain name and served by the same web server; a browser is the client program that requests and displays pages. Email, file transfer and online gaming also use the Internet, but they are not part of the Web.

IP addresses, domain names and DNS

Every device on the Internet must have a unique address, its IP address. A dynamic address is assigned each time a device connects, while a static address stays the same, which suits servers. IPv4 addresses are 32 bits long, giving roughly 4.3 billion possibilities, and are written as four 8-bit octets in decimal separated by full stops, such as 86.47.193.12. The growth of the Internet has used up IPv4 addresses, so IPv6 uses 128 bits, about 3.4 × 10^38 addresses, written as eight groups of four hexadecimal digits separated by colons. Because numbers are hard to remember, people use domain names instead. A domain name is read from right to left: the final label is the top-level domain, such as .uk or .com, and each label to the left is a sub-domain of the part after it, so www.belfastzoo.co.uk is a sub-domain of belfastzoo.co.uk, which is part of co.uk. The Domain Name System translates names into IP addresses. It is a distributed hierarchy of servers that pass queries along and exchange updates, so a name still leads to the right device when its address changes, and answers are cached to speed up later look-ups.

Connecting, hosting and URLs

Homes and businesses reach the Internet through an internet service provider, which connects them to a Point of Presence, its access point for a geographical area. A router, often combined with a modem, shares one connection between many devices. Many ISPs also offer email, domain name registration and web hosting. To publish a website, an organisation rents space from a web host, whose package normally includes a fixed amount of storage, a limit on traffic, email addresses, domain registration, database support, backups and traffic statistics. Pages are uploaded to the host using the File Transfer Protocol, which authenticates the user with a username and password and can move files of any type in either direction. Anonymous FTP, with no login, is used only to make files publicly downloadable. Each resource on the Web is identified by a uniform resource locator. A URL combines the protocol, the domain name of the server and the path to the file, for example https://www.translink.co.uk/timetables/metro.html, and may finish with a query string carrying data. In general, pages are uploaded to the server by FTP and then requested by visitors' browsers using HTTP or HTTPS.

HTTP, cookies and sessions

The HyperText Transfer Protocol is the set of rules a browser uses to request web pages and a server uses to reply. It is stateless: every request is a separate transaction with no memory of any earlier request. This makes the protocol simple to implement and efficient, the two main advantages, but it causes problems whenever a site needs context. An online bank, for instance, must know that the person asking to transfer money is the same person who logged in a moment ago. Two techniques solve this. A cookie is a small piece of text that the site asks the browser to store; the browser then sends it back with each request, so the server can recognise the user, remember preferences or keep track of a shopping basket. A session variable is stored on the server for the duration of the visit, and the browser holds only a session identifier, usually inside a cookie. Sessions are more secure because the sensitive data never sits on the user's device. Cookies also raise privacy concerns, since tracking cookies can follow users across sites. In the UK, the Privacy and Electronic Communications Regulations, together with the UK GDPR, require consent before non-essential cookies are stored.

HTTPS, caching and the page request

When a URL is entered, the browser first looks in its cache for the page and the server's IP address. If the address is not known, a DNS resolver finds it by querying the DNS hierarchy. The browser then connects to the web server and sends an HTTP GET request; for HTTPS a TLS connection is set up first, the server proves its identity with a digital certificate and all data is encrypted. The server returns a status code such as 200 OK or 404 Not Found together with the HTML. The browser then requests the style sheets, scripts and images that the page refers to and renders the result. Because HTTP sends every page in full, items common to many pages, such as logos and menus, would be downloaded repeatedly. Caching reduces this. A browser cache on the device stores items that user has already fetched, and a proxy cache on a shared server, for example in a school, stores content that many users request. Requests are checked against the cache before going to the web server, reducing traffic, waiting time and cost. Pages whose content changes rapidly, such as live sports scores, can tell browsers not to cache them, or only for a short time.

Search engines

In the Web's early years, lists of servers were kept by hand, but the number of sites soon made that impossible, and automatic search engines appeared from 1994. A modern search engine has three main parts. The web crawler, sometimes called a spider, starts from a set of seed pages, collects every link on them, visits each linked page in turn and adds any new links to its list, aiming to reach as much of the Web as possible and returning regularly to spot changes. The index stores what the crawler finds on every page, including its text, images, video and metadata, in a dictionary-like structure so that each word can be looked up to find the pages containing it. Metadata is data about the page, such as its title and description, that is not part of the visible content. The query engine is the part the user sees: it takes the search terms, looks them up in the index and ranks the matching pages so the most useful appear first. Google rose to dominance around 2000 with its PageRank algorithm, which scores a page more highly when many other pages, particularly highly ranked ones, link to it. Modern engines combine many other ranking signals as well.

Web standards bodies

The Internet and the Web work only because devices, networks and browsers made by thousands of organisations follow the same rules. Several standards bodies agree these rules. The Internet Engineering Task Force is an open, largely volunteer organisation that develops the core Internet protocols, including IPv4, IPv6, email and the Domain Name System. The International Telecommunication Union is a United Nations agency responsible for telecommunications. Its standardisation sector, ITU-T, specifies how data is carried and exchanged between devices: transmission over cables, radio and satellite links, the coding of digital video, sound and images, and the format of the digital certificates used to secure e-commerce. The World Wide Web Consortium, founded by Tim Berners-Lee, brings together member organisations to agree the standards for web content and web applications, such as HTML, CSS and accessibility guidelines. Its standards mean a web page should look and behave the same in any browser on any device. In short, the ITU deals mainly with how data is formatted and transmitted, while the W3C deals with how information to be displayed on web pages is specified. Without such agreement, browsers, servers and networks from different makers could not work together.

T Key terms from the mark schemes

Internet
A global network of interconnected computer networks that use TCP/IP so that any connected device can communicate with any other.

World Wide Web
A collection of linked web pages and resources, each with a unique address, accessed over the Internet using HTTP and browsers.
`Credited: WWW`

IP address
A unique numeric address assigned to a device on a network such as the Internet: 32 bits in IPv4 and 128 bits in IPv6.

Domain Name System (DNS)
A distributed, hierarchical system of servers that translates domain names into IP addresses.

Top-level domain (TLD)
The last part of a domain name, such as .uk, .ie or .com, which may indicate the country or type of organisation.

URL
Uniform resource locator: the full address of a resource on the Web, made up of the protocol, domain name and path or file name.
`Credited: Web address`

Client–server model
A model in which client devices send requests to a central server, which processes them and returns responses.

Web server
A computer running software that stores web pages and delivers them to browsers when requested using HTTP.

Internet service provider (ISP)
A company that provides users with a connection to the Internet, often with services such as email and web hosting.

HTTPS
Hypertext Transfer Protocol Secure: HTTP sent over an encrypted TLS connection, with the server authenticated by a digital certificate.
`Credited: HTTP over SSL/TLS`

Cookie
A small piece of text data sent by a website and stored by the browser, then returned with each request so the site can remember information about the user.

Session
Data about a user's visit stored on the web server and identified by a session ID held by the browser.
`Credited: Session variable`

Cache
Temporary storage of previously retrieved web content, in the browser or on a proxy server, checked before requesting content from the web server.

File Transfer Protocol (FTP)
A standard protocol for transferring files between computers, commonly used to upload website files to a web host.

Exam focus

Definition questions on terms such as web server, ISP, cache or cookie usually give one mark for what the thing is and one for what it does or why it is used, so include both. When asked how the Internet and Web differ, say the Internet is the hardware network of networks and the Web is a service of linked pages accessed over it. Sequence questions on loading a page reward the correct order: DNS look-up of the IP address, HTTP request to the web server, response and rendering. Say that DNS converts a domain name into an IP address, not the reverse. For HTTP versus FTP, link FTP to uploading files to a server and HTTP to a browser requesting pages. Questions on statelessness expect both advantages, simplicity and efficiency, and an explanation of how cookies or session variables keep track of a logged-in user. To explain sub-domains, read the name from right to left. Learn the three search engine components with what each does. For standards, link the W3C with web content languages such as HTML and CSS and the ITU with communications and data formats. Common errors are calling DNS 'Domain Name Server' when the system is meant, and claiming IPv4 addresses are 128 bits long.

? Check yourself

1. Explain the difference between the Internet and the World Wide Web. [2]
2. Explain the meaning of each of the terms web page, website and web server, and describe the relationship between them. [4]
3. The IPv4 address of a council's web server is written in binary as 11000011 00010110 00001010 01000001. Convert it to dotted decimal notation, showing your working. [3]

Answers: 1. The Internet is a global network of interconnected networks / the physical infrastructure and protocols (1) / the World Wide Web is a service running over the Internet: a collection of linked web pages accessed with a browser (1) 2. Web page: a single hypertext document viewed in a browser (1) / website: a set of related web pages with a common domain name (1) / web server: computer/software that stores web pages and delivers them when requested by a browser using HTTP (1) / relationship: a website is made up of web pages, stored on and delivered by a web server, and together all websites form the WWW (1) 3. 11000011 = 128 + 64 + 2 + 1 = 195 / 00010110 = 16 + 4 + 2 = 22 / 00001010 = 8 + 2 = 10 / 01000001 = 64 + 1 = 65 (1 for working shown for at least two octets, 1 for at least three octets correct) / final answer 195.22.10.65 (1)